DATA SCIENCE CAPSTONE

SpaceX Falcon 9

First-Stage Landing Prediction

A technical story from data collection to decision support

- Data collection & wrangling
- EDA, SQL, geography & dashboarding
- Classification and model evaluation

Final presentation • October 2026

Can we estimate recovery success?

Use historical launch characteristics to support launch-risk and bid planning.

90
launch records

66.7%
successful landing share

83.3%
published course-style benchmark*

- The landing label is imbalanced: 60 successes versus 30 unsuccessful outcomes.
- Reported site-level rates vary: KSC LC-39A and VAFB SLC-4E are near 77%, while CCAFS SLC-40 is about 60%.
- A small 18-record test set makes accuracy unstable; treat the benchmark as a reference, not a guarantee.

*83.3% is a published course-style reference (15/18 correct), not a fresh model run in this document.

Why first-stage recovery matters

Reusable boosters can reduce launch costs, but recovery outcomes depend on mission and operational factors.

- Business framing: estimate whether a Falcon 9 first stage lands successfully.
- Potential users: a competing launch provider assessing bid risk, mission profiles and cost assumptions.
- Target variable: Class = 1 for successful landing; Class = 0 for unsuccessful landing.
- Predictors explored: launch site, orbit, payload mass, flight number and booster-related fields.
- Scope: historical classification—not a causal explanation or a guarantee of future performance.

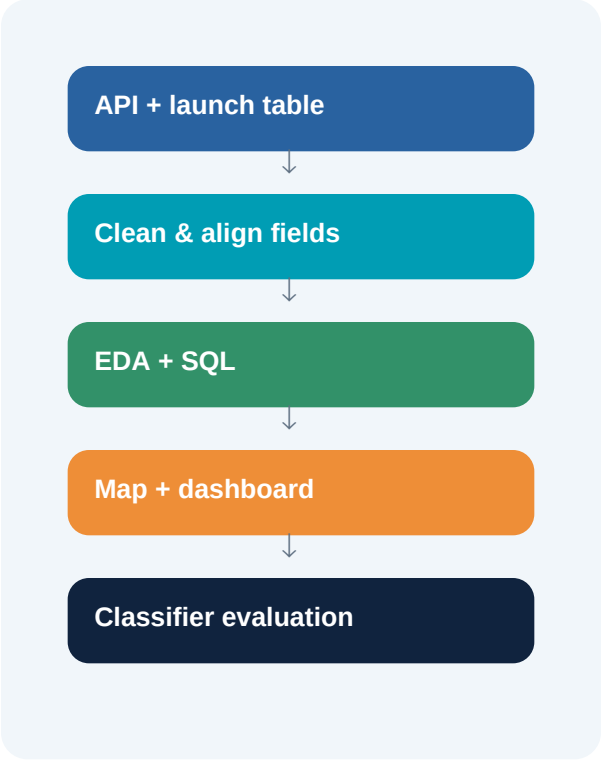
PROJECT QUESTION

How well can launch characteristics help distinguish successful from unsuccessful first-stage landings?

Combine structured launch records

The course workflow uses SpaceX API information and launch-table data, then prepares a consistent launch-level table.

- Source dataset: IBM/Coursera cleaned file dataset_part_2.csv.
- Working table: 90 rows × 18 columns in the standard course version.
- Fields include flight number, date, payload mass, orbit, launch site, outcome, booster information, coordinates and Class.
- Cross-check dates, categories and target labels before analysis.
- Keep the data version and definitions consistent across EDA, SQL, mapping and classification.



Turn raw records into a modeling table

Data quality decisions matter because missing or inconsistent labels can distort the measured success rate.

- Inspect column types, missing values, duplicates and category consistency.
- Keep relevant Falcon 9 records and verify one record per launch for this analysis.
- Use Class (1 = success, 0 = unsuccessful); do not silently convert unknown outcomes to failures.
- LandingPad is missing for roughly 28.9% of records in the standard 90-row dataset.
- For modeling, encode categories and fit imputation/scaling inside a training-only pipeline.

DATA QUALITY CHECKLIST

Schema	18 fields
Rows	90 launches
Target	binary Class
Missingness	LandingPad ~28.9%
Leakage control	train-only preprocessing

Explore outcomes before modeling

Test whether observed patterns vary by launch site, mission type, payload and flight experience.

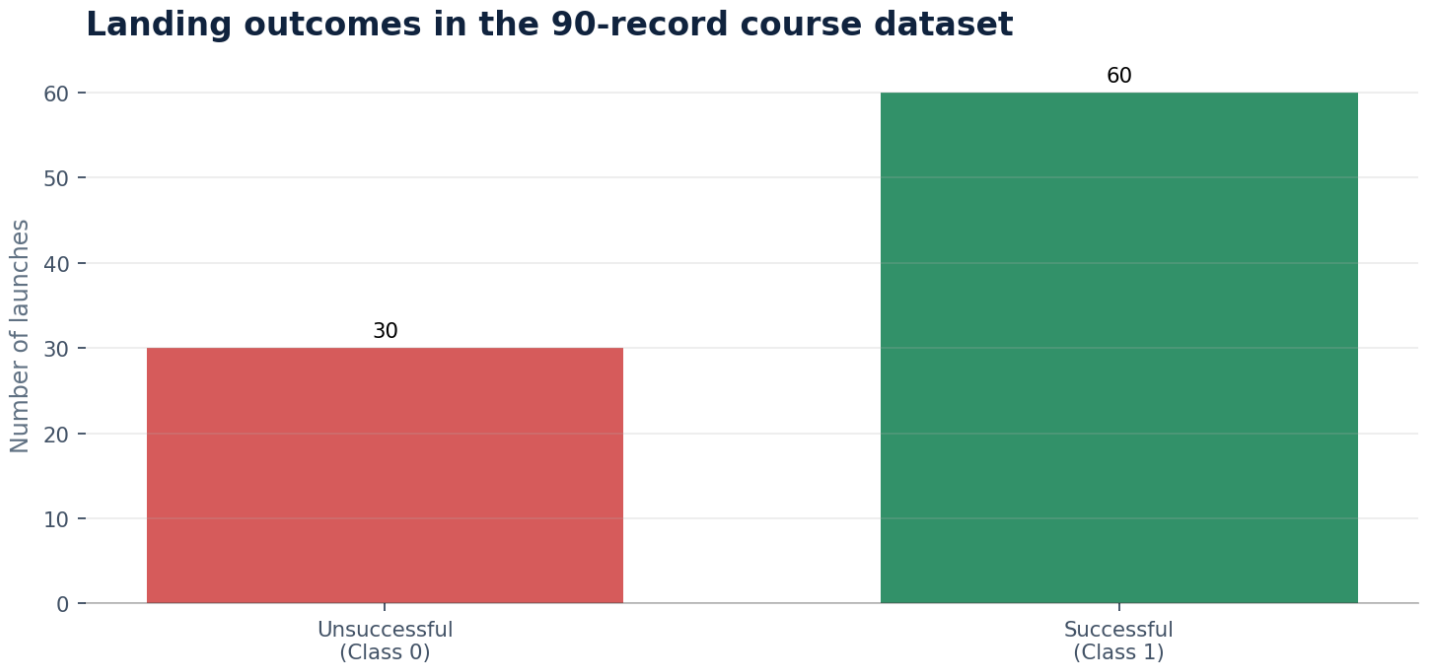
- Univariate analysis: count successful and unsuccessful outcomes; inspect payload distributions.
- Grouped analysis: compare success proportion by launch site and orbit.
- Trend analysis: inspect outcome patterns over flight number and launch year.
- SQL analysis: reproduce counts and group-level rates with transparent queries.
- Interpretation rule: show group size alongside rates; small samples can produce extreme percentages.

ANALYTIC QUESTIONS

- Is the target balanced?
- Which sites have more launches?
- Do success rates differ by orbit?
- Does performance improve over time?
- Which predictors may help classification?

Landing outcomes: 2 in 3 were successful

The standard course dataset contains 90 launches: 60 successes and 30 unsuccessful outcomes.



KEY FINDINGS

60 / 90

successful outcomes (66.7%)

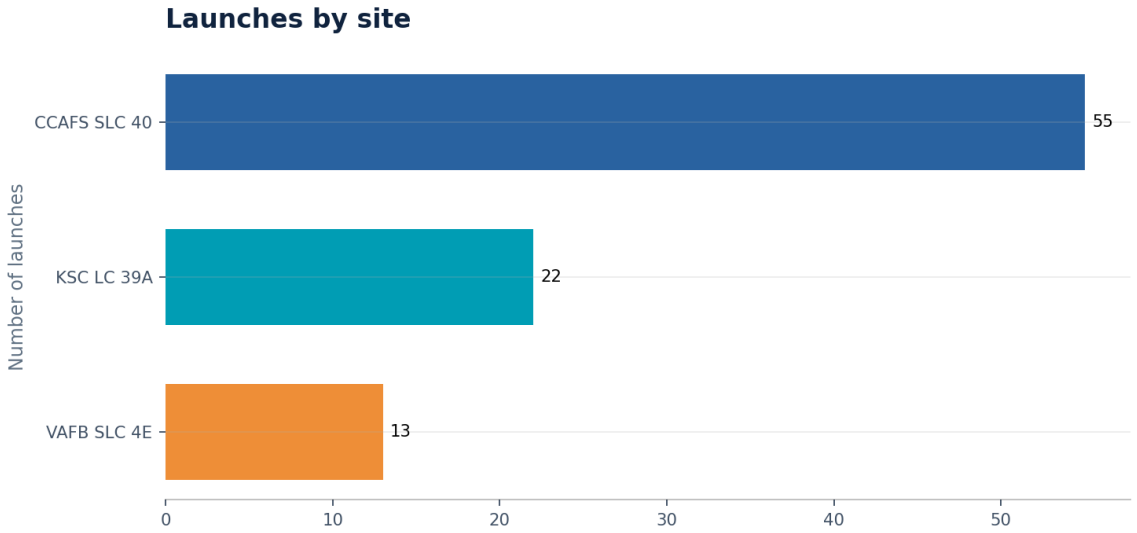
30 / 90

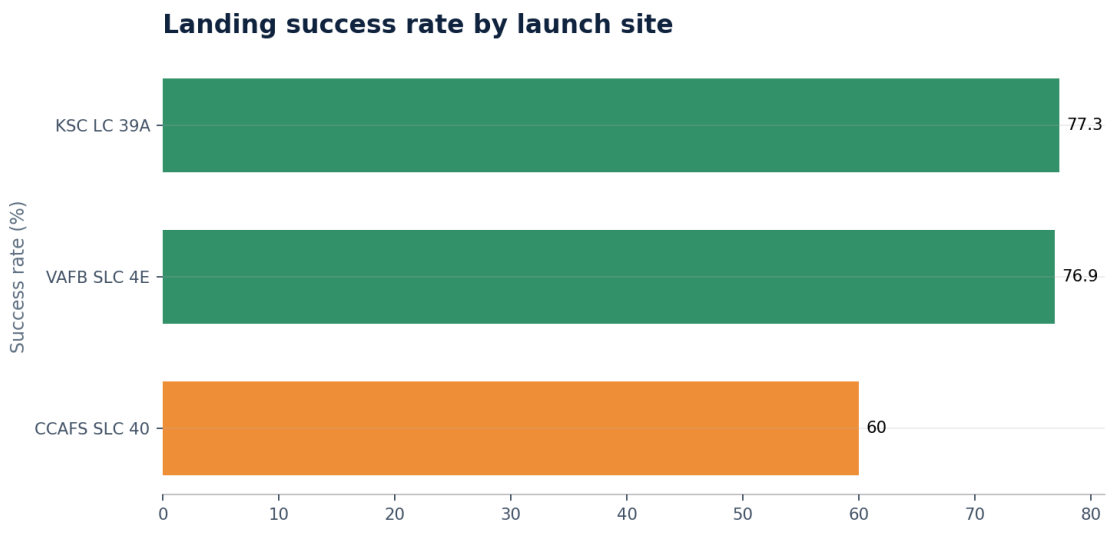
unsuccessful outcomes (33.3%)

Evaluation should consider precision, recall and the confusion matrix—not accuracy alone.

Launch-site patterns differ

Counts and rates reported for the standard course dataset, rounded to one decimal place.



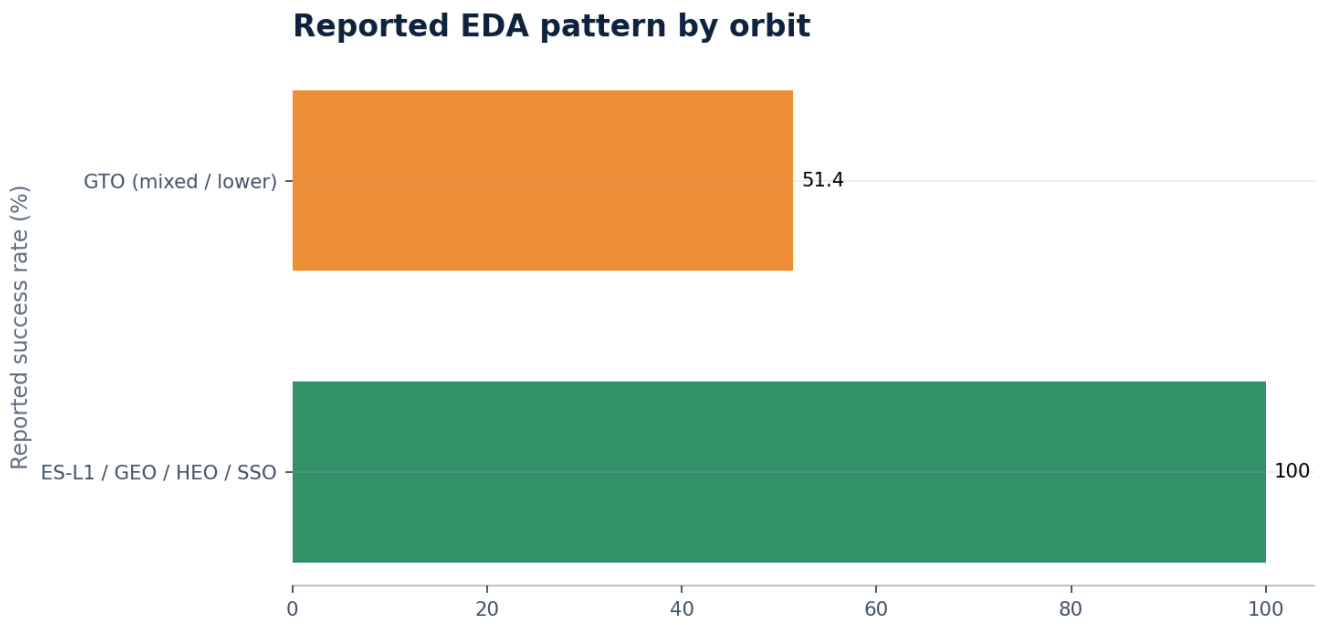


Site	Launches	Success rate	Interpretation
CCAFS SLC 40	55	60.0%	Largest group; lower rate

KSC LC-39A and VAFB SLC-4E have higher reported rates, but group sizes are smaller.

Orbit-level rates require sample-size caution

Public course-style EDA reports very high rates for some low-frequency orbits and mixed outcomes for GTO.



INTERPRETATION

- ES-L1, GEO, HEO and SSO are reported at 100% in this course dataset.
- These orbit categories have few launches; observed 100% success is not a guarantee.
- GTO shows more mixed outcomes in published course analyses.
- Orbit should be evaluated with site, payload and flight history.

Rates are descriptive and may change with dataset version or inclusion of additional launches.

SQL makes comparisons auditable

The results summarize the standard course table; query patterns can be reproduced with GROUP BY and AVG(Class).

SQL check	Result	Meaning
COUNT(*)	90 launches	Dataset row count
COUNT(Class = 1)	60	Successful landings
COUNT(Class = 0)	30	Unsuccessful outcomes
AVG(Class)	0.6667	Overall success proportion
GROUP BY LaunchSite	55 / 22 / 13	CCAFS / KSC / VAFB launches

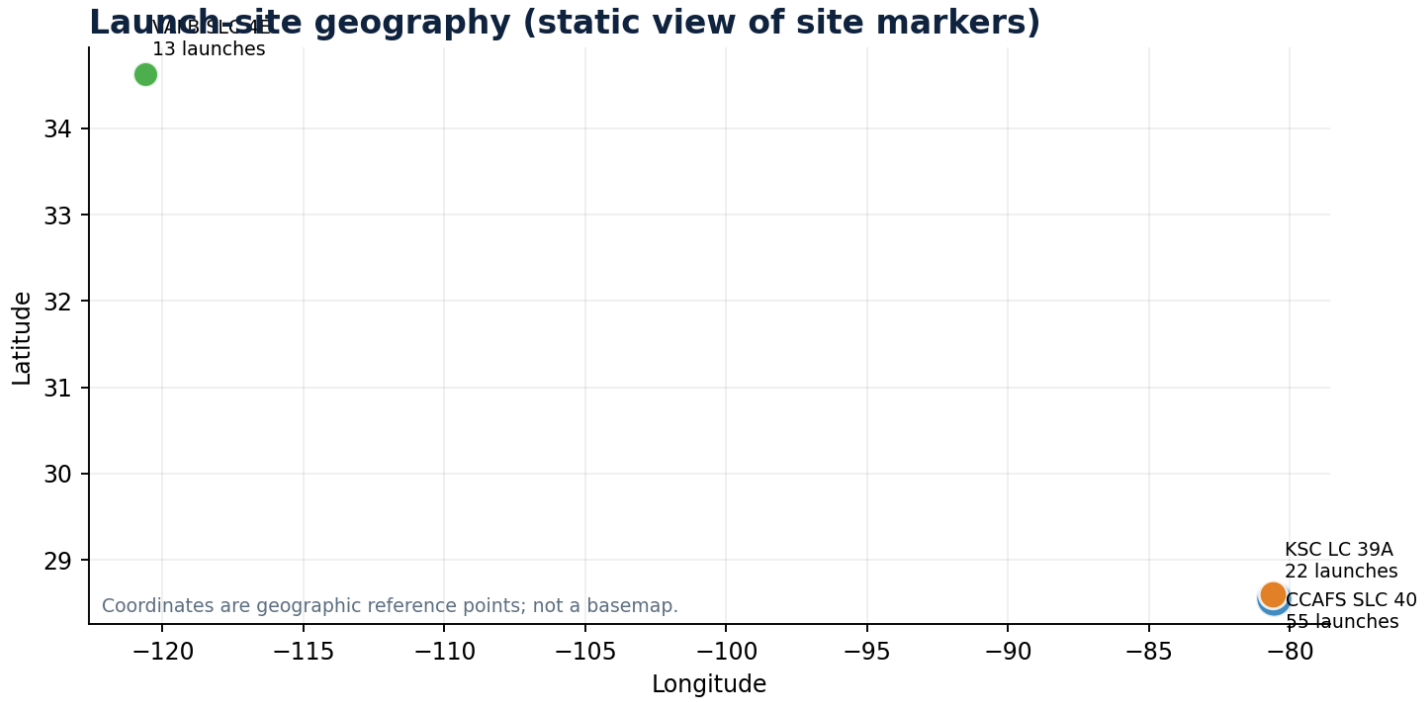
Example SQL

```
SELECT LaunchSite, COUNT(*) AS launches, AVG(Class) AS success_rate
FROM SPACEX GROUP BY LaunchSite;
```

AVG(Class) is a success rate only when Class is consistently encoded 0/1 and missing targets are excluded.

Folium: launch-site geography

The course lab uses markers and popups; the image below is a static geographic preview for this PDF.



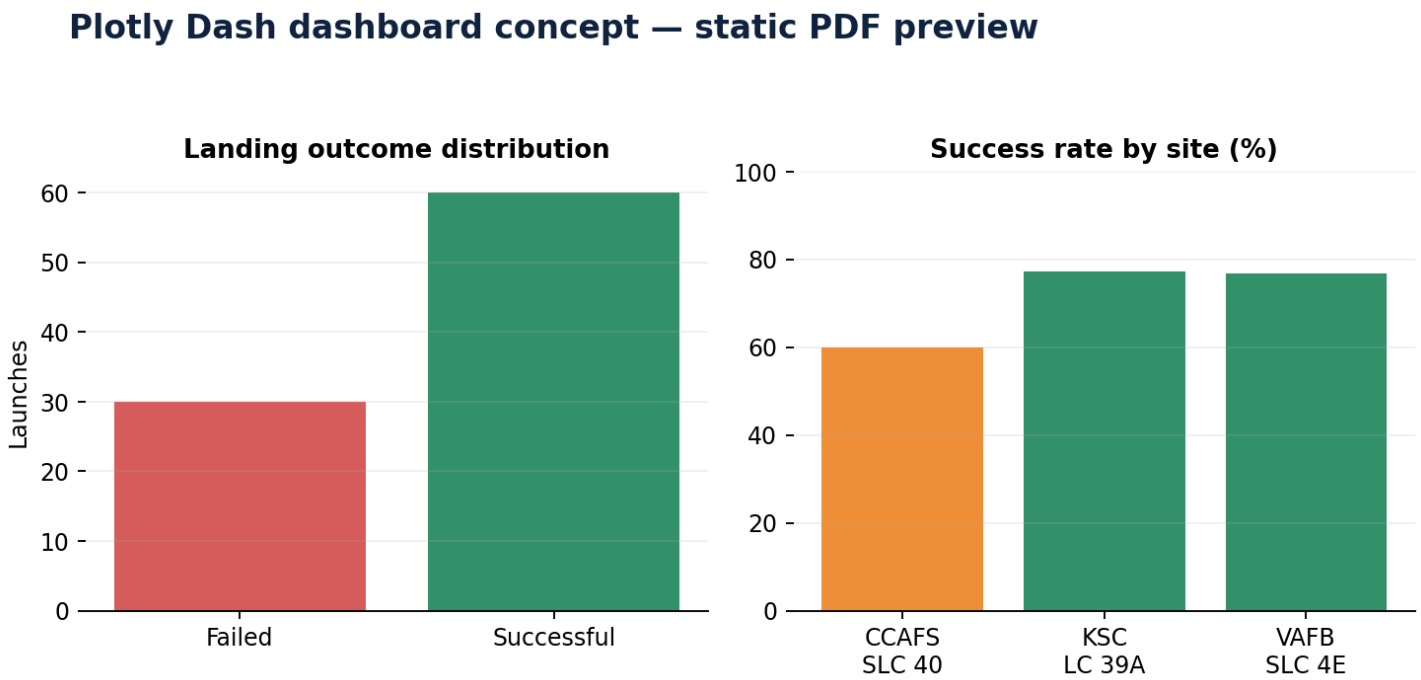
GEOGRAPHIC OBSERVATIONS

- Two sites are in Florida and one is in California.
- All are coastal launch facilities, consistent with over-water trajectory and safety considerations.
- The map is descriptive; geography alone does not establish causation.

Coordinates: CCAFS SLC 40 (28.562°N, 80.577°W); KSC LC-39A (~28.608°N, 80.604°W); VAFB SLC-4E (34.632°N, 120.611°W).

Plotly Dash: from analysis to exploration

A dashboard lets stakeholders filter site and payload subsets and compare outcomes without editing code.



DASHBOARD COMPONENTS

- Launch-site filter
- Payload-mass range selector
- Outcome distribution chart
- Success rate by site
- Payload vs. landing outcome scatter plot

Static preview uses reported summary values; it is not a live running Dash app screenshot.

Classification methodology

Compare models on the same held-out test set, using training-only preprocessing to reduce leakage.

- Target: Class (1 = successful landing, 0 = unsuccessful landing).
- Candidate classifiers: Logistic Regression, Support Vector Machine, Decision Tree and K-Nearest Neighbors.
- Encode categorical predictors; impute missing values and scale numeric variables where needed.
- Use a stratified train/test split; perform tuning with cross-validation on training data only.
- Report accuracy, precision, recall, F1-score and confusion matrix; prefer risk-aware evaluation.

EVALUATION METRICS

Accuracy
overall fraction correct

Precision
reliability of predicted successes

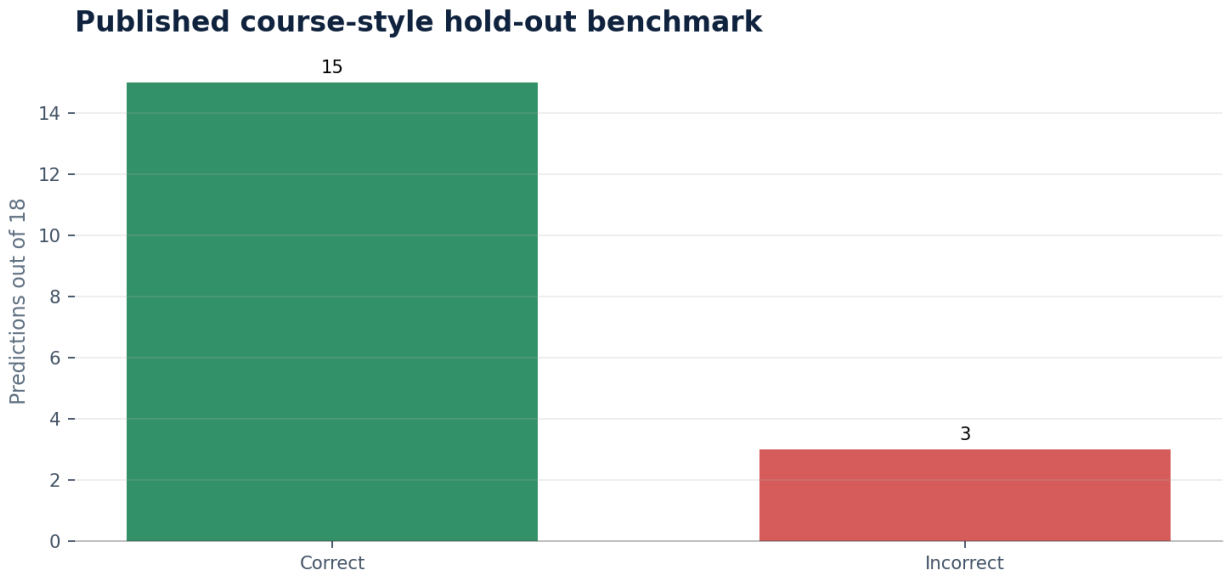
Recall
successes detected

F1
precision/recall balance

Confusion matrix
false positives and negatives

Reference benchmark: 15 of 18 correct

A course-style public reference reports 83.33% test accuracy; this PDF does not claim an independent rerun.



REFERENCE CONFUSION MATRIX



Published illustrative reference; exact matrix depends on the positive class and split.

With 18 test rows, one changed prediction moves accuracy by 5.56 percentage points.

Move from a score to a decision

A useful model should communicate uncertainty and help decision-makers compare risk—not simply output yes/no.

INSIGHTS TO USE

- Landing success improved across flight history in published course analyses.
- Site and orbit differences are entangled with mission mix and launch era.
- Use predicted probability and the cost of false success predictions to support bids.
- Show sample sizes next to percentages, especially for rare orbits.

LIMITATIONS & NEXT STEPS

- Small historical dataset and changing launch technology.
- Missing landing-pad metadata and possible class imbalance.
- Metrics vary by feature set, random split and preprocessing.
- Rerun on the exact course CSV; use repeated stratified CV and probability calibration.

A reproducible workflow supports better decisions

The capstone connects data preparation, EDA, SQL, geography, dashboards and classification.

- In the standard 90-record course dataset, 60 landings are successful and 30 are unsuccessful (66.7% success).
- Reported site rates range from about 60% at CCAFS SLC 40 to about 77% at KSC LC-39A and VAFB SLC-4E.
- A published course-style classification benchmark reports about 83.3% accuracy on an 18-record test set; it is not independently reproduced here.
- Operational use requires a fresh run on the exact dataset, model metrics, confusion matrix and live Folium/Dash outputs.

SOURCES

IBM Skills Network course dataset: dataset_part_2.csv

Course notebook reference: sentinel-1.github.io/IBM-Applied-Data-Science-Capstone/

Public course-style benchmark and site patterns: ResearchGate, IBM Data Science Professional Certificate Capstone (2024)

Note: GitHub URL intentionally omitted per request. The rubric you shared lists it as a 1-point item, so omission may affect the score.